ESS SENTINEL 2.0

Scenario report

2026-09-29T09:25:30.612465+00:00

Source: Synthetic reference dataset

Batch: batch-732c43b47f014f6e

Model: ESS-2026-09-29T050332-ebf09b1e

Screening stage: 24h

Decision-support prototype. Synthetic results are not real-world qualification evidence. Final disposition requires qualified engineering review.

Scenario comparison

population	components	review	static violations	predicted crossings
Baseline	120	25	0	0
Latent defect	120	26	0	1

Methodology and limitations

Robust peer statistics, early-feature Isolation Forest and lot-disjoint regression support transparent review rules. Scores are not failure probabilities. Calibration intervals are approximate under lot dependence. Synthetic evidence is not production validation. Review states do not alter engineering measurements or decisions.